



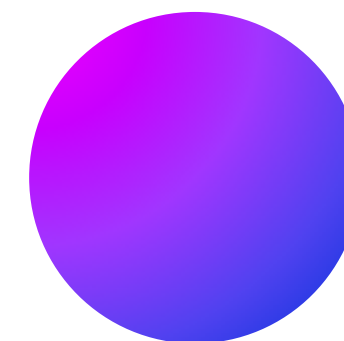
Télécom IA

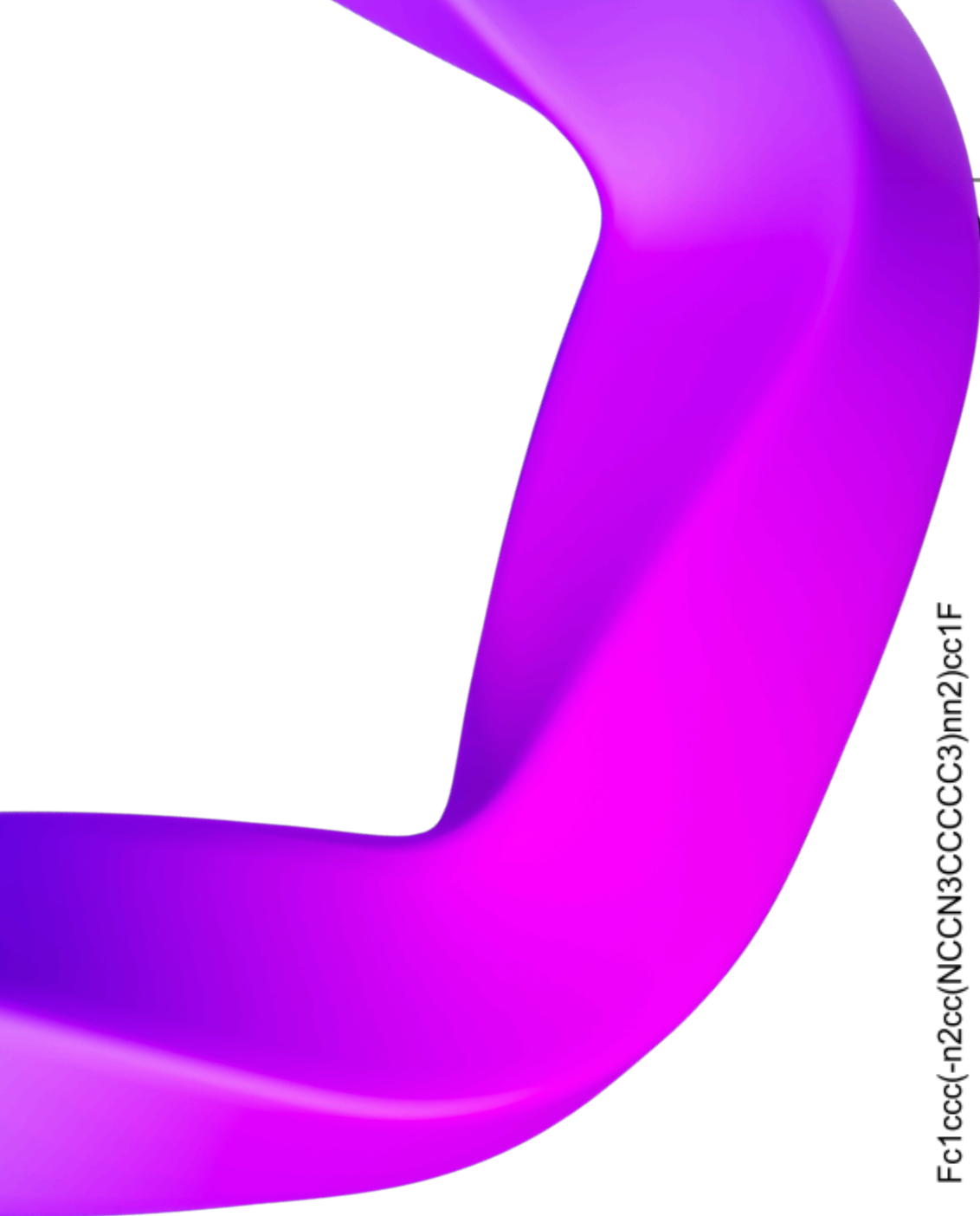


hERG toxicity prediction

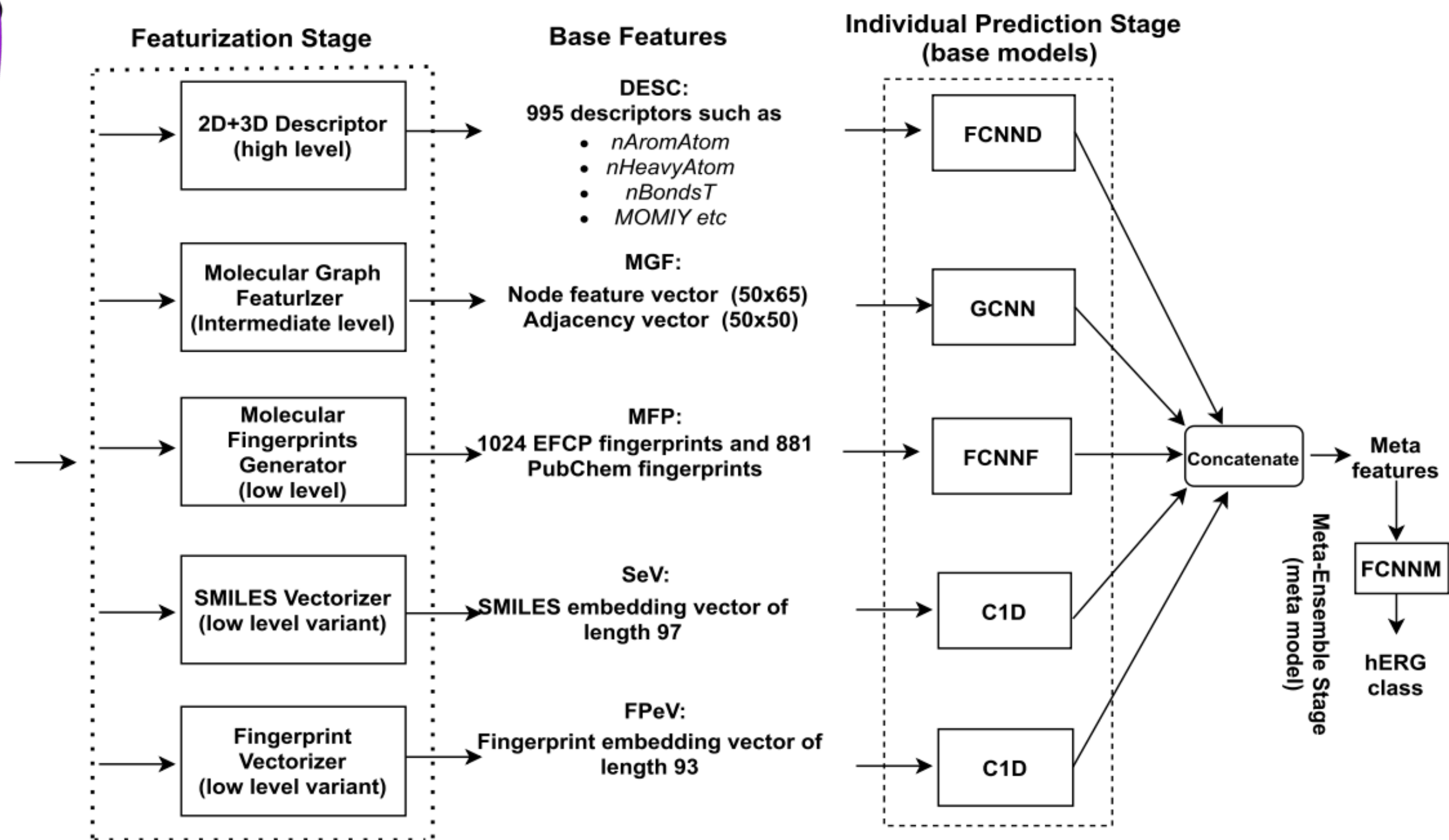
Final pitch

Noa A. - Alexandre M. - Rayane D.

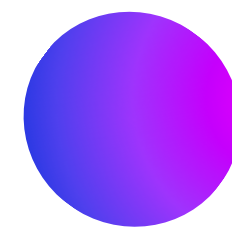




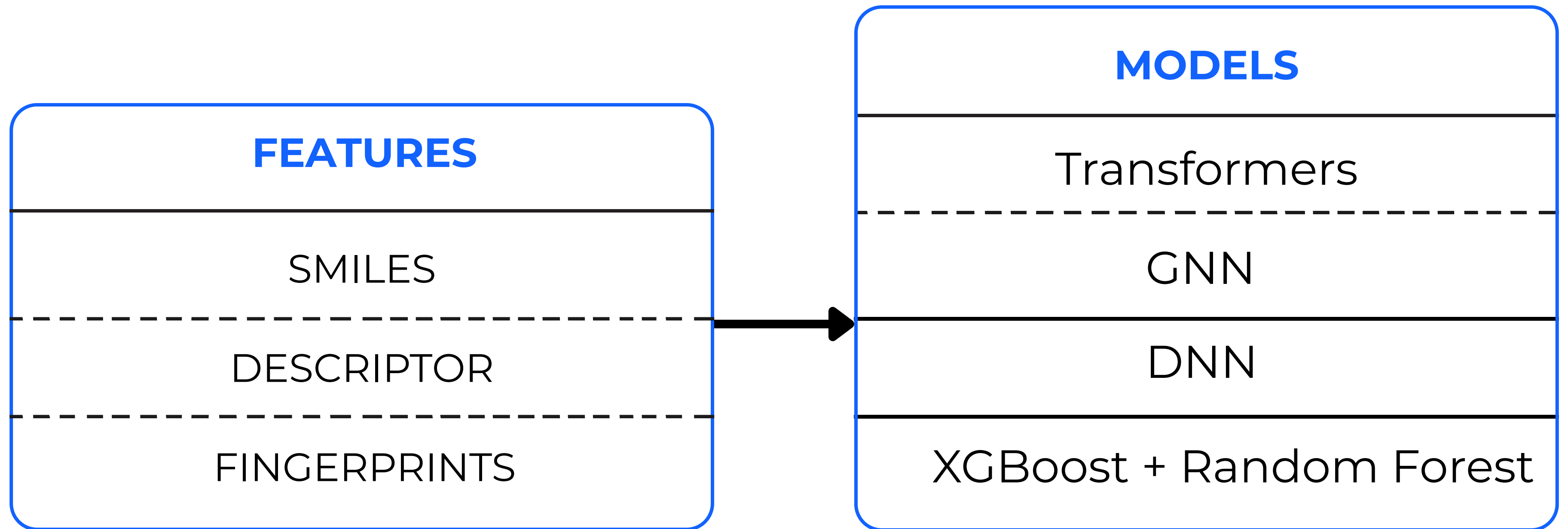
Fc1ccc(-n2cc(NCCN3CCCCC3)nn2)cc1F



CardioTox net:



Notre Idée

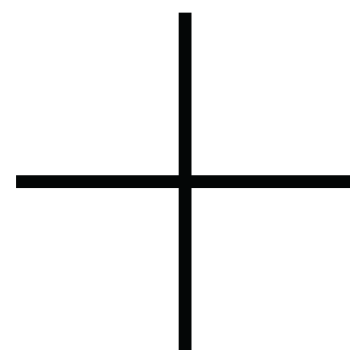


Des Transformers ?

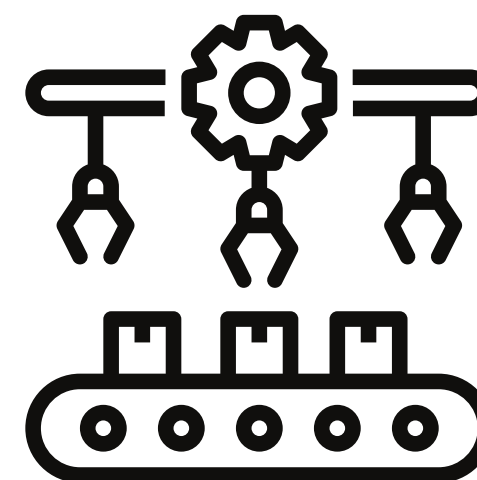


seyonec/ChemBERTa-zinc-base-v1

Pour représentations chimiques de
molécules au format SMILES

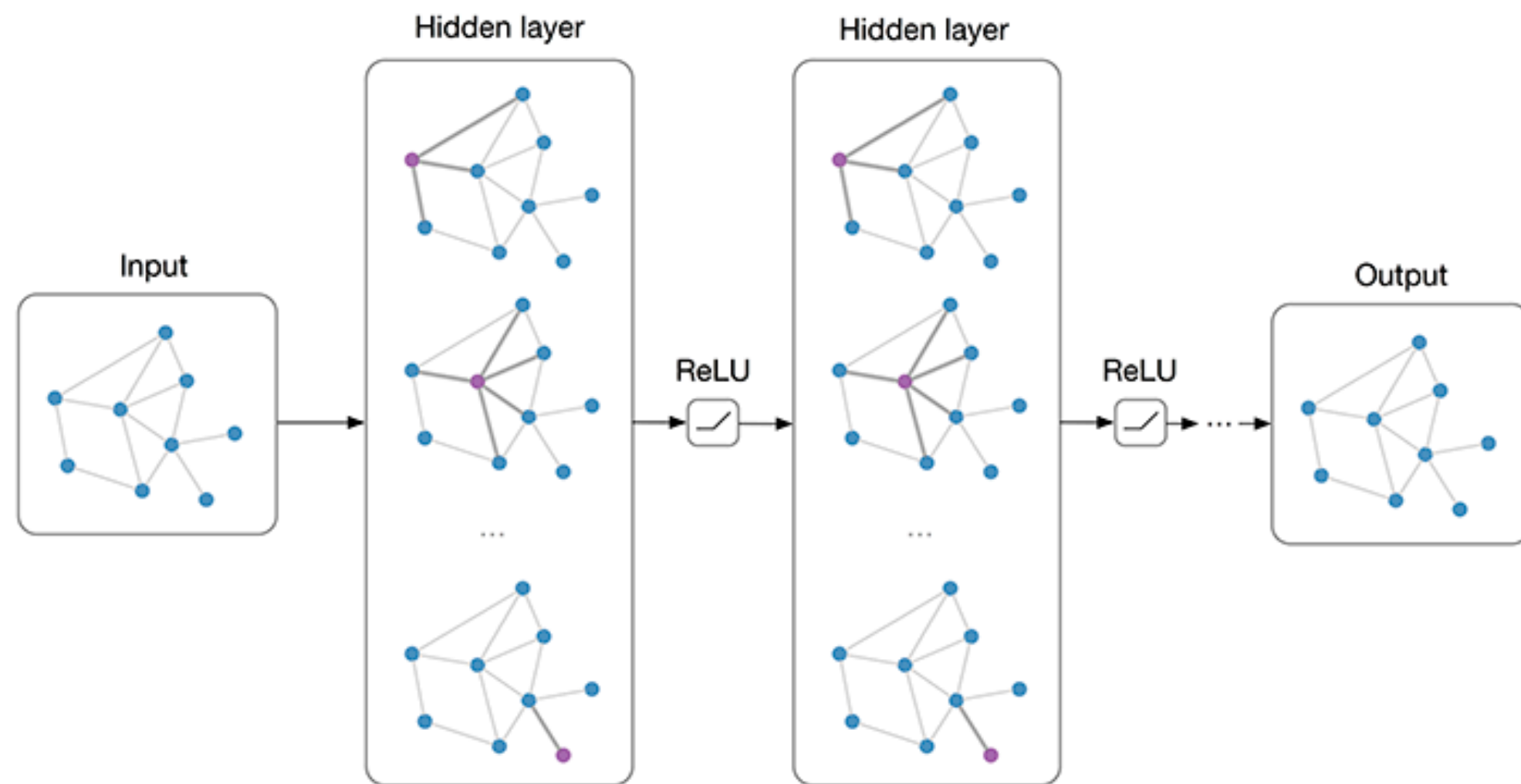


FINETUNING



Graph Convolutional Neural Network

Construction
des graphes
avec RDKit

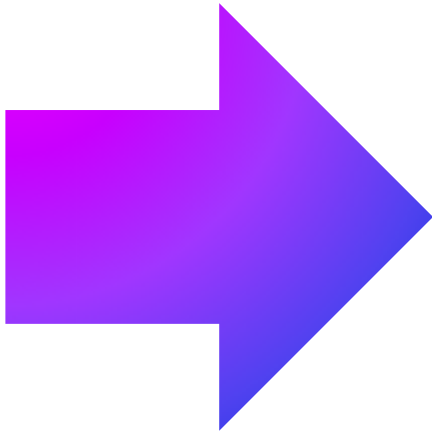


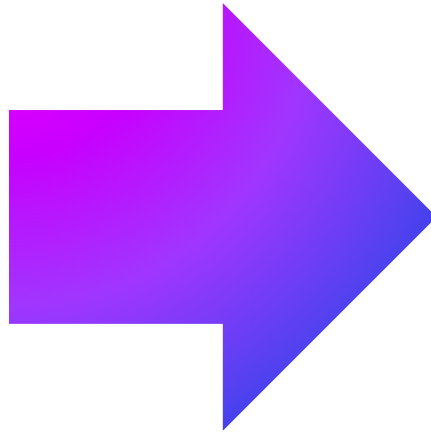
$$L_{i,j} := \begin{cases} \deg(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \text{ is adjacent to } v_j \\ 0 & \text{otherwise,} \end{cases}$$

TASK 1 and 3

Meta modèle

Probabilités de chaque modèle :

- 
- Neural Network
 - Moyenne pondérée
 - Vote de majorité
 - Logistic Regression



Task 1 : Classification

Task 3 : Ordre des probas

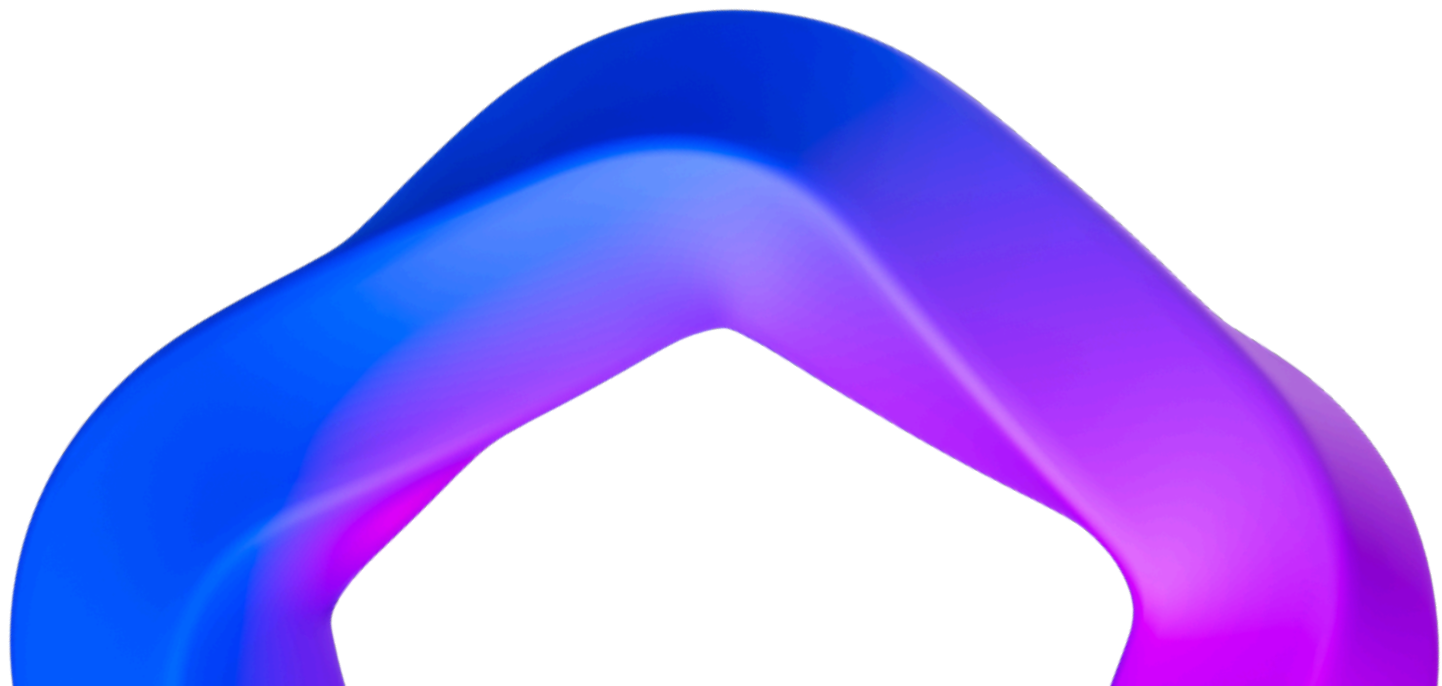
TASK 2

Task 2

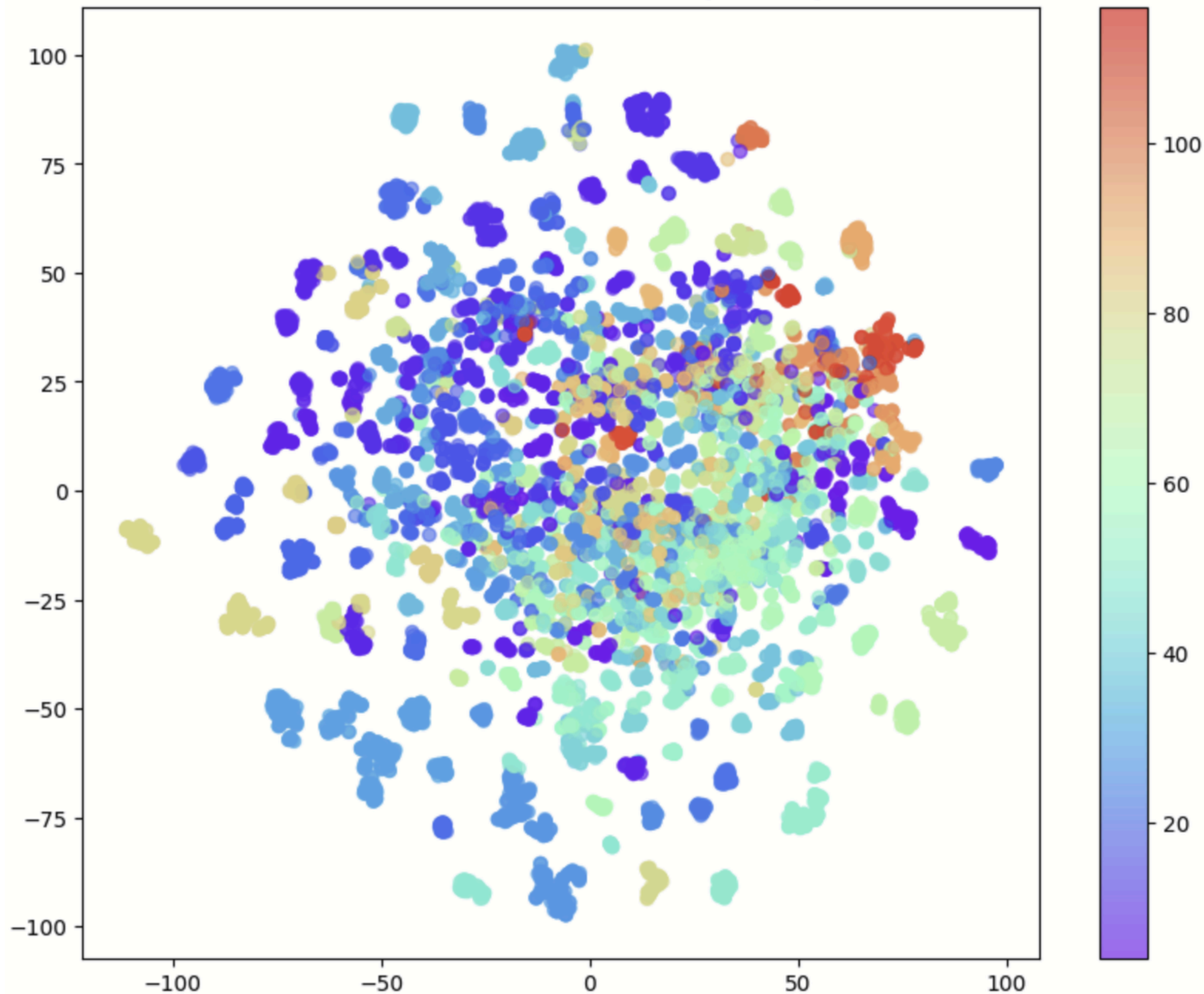
Vector ecfc + fcfc



**Matrice de
distance de
Tanimoto**



Visualisation des Clusters avec {method}



Nombre de clusters hiérarchiques: 89

Taille des plus grands clusters:

Cluster 1: 430 molécules

Cluster 2: 404 molécules

Cluster 3: 373 molécules

Cluster 4: 368 molécules

Cluster 5: 283 molécules

Cluster 6: 271 molécules

Cluster 7: 270 molécules

Cluster 8: 238 molécules

Cluster 9: 221 molécules

Cluster 10: 216 molécules

Cluster 11: 213 molécules

Cluster 12: 210 molécules

Cluster 13: 195 molécules

Cluster 14: 169 molécules

Cluster 15: 168 molécules

Cluster 16: 158 molécules

Cluster 17: 158 molécules

Cluster 18: 144 molécules

Cluster 19: 137 molécules

Cluster 20: 130 molécules

Taille moyenne des clusters: 80.47008547008546

Deviation standard des tailles de clusters: 87.66218889

linkage + fcluster

```
rf = RandomForestClassifier(random_state=42, n_jobs=-1)

param_grid = {'max_depth': [20],
              'max_features': ['sqrt'],
              'min_samples_leaf': [1],
              'min_samples_split': [5],
              'n_estimators': [300]
              }

cluster_train_labels = X_train['cluster'].values
cv = GroupKFold(n_splits=5).split(X_train.drop(columns=['cluster']), y_train, groups=cluster_train_labels)
```

- Analyse des distributions
- Accuracy par serie
- Cross-Validation “serie-aware”
- Grid Search pour les meilleurs paramètres



MERCI